

	Predicted Class			
Actual class		Yes	No	
	Yes	2750	250	3000
	No	150	2850	3000
		2900	3100	6000

Confusion matrix

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{2750}{2750 + 250}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{FP} + \text{TN}} = \frac{2850}{2850 + 150}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{2750}{2750 + 150}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{2750}{2750 + 250}$$

		Predicted Class		
		Class A	Class B	Class C
Actual Class	Class A	900	75	25
	Class B	100	800	100
	Class C	50	100	850

Sensitivity (Class A) = $900/1000$

Specificity (Class A) = $(800+850)/(1000+1000)$

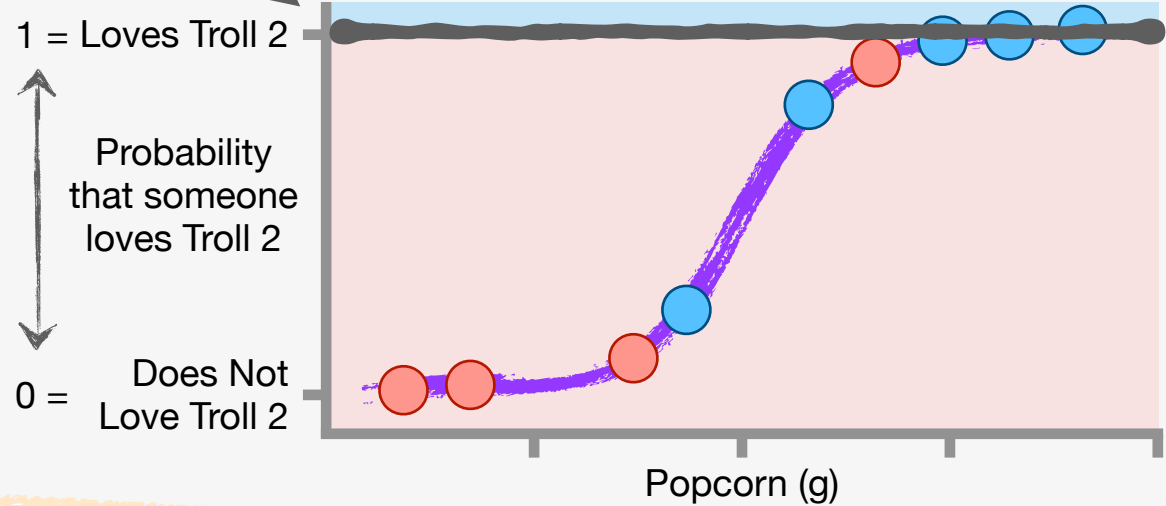
Precision (Class A) = $900/(900+100+50)$

Recall (Class A) = $900/1000$

Find out the same for other classes

ROC: Details Part 1

1 To get a better sense of how an **ROC** graph works, let's draw one from start to finish. We'll start by using a classification threshold, **1**, that classifies everyone as someone who **Does Not Love Troll 2**...



Gentle Reminder:

	Predicted		
	Yes	No	
Actual	Yes	TP	FN
	No	FP	TN

True Positive (arrow to TP), False Negative (arrow to FN), False Positive (arrow to FP), True Negative (arrow to TN)

...and when the classification threshold is set to **1**, we end up with this **Confusion Matrix**.

	Predicted	
	Yes	No
Actual	Yes	0
	No	4

Threshold = 1

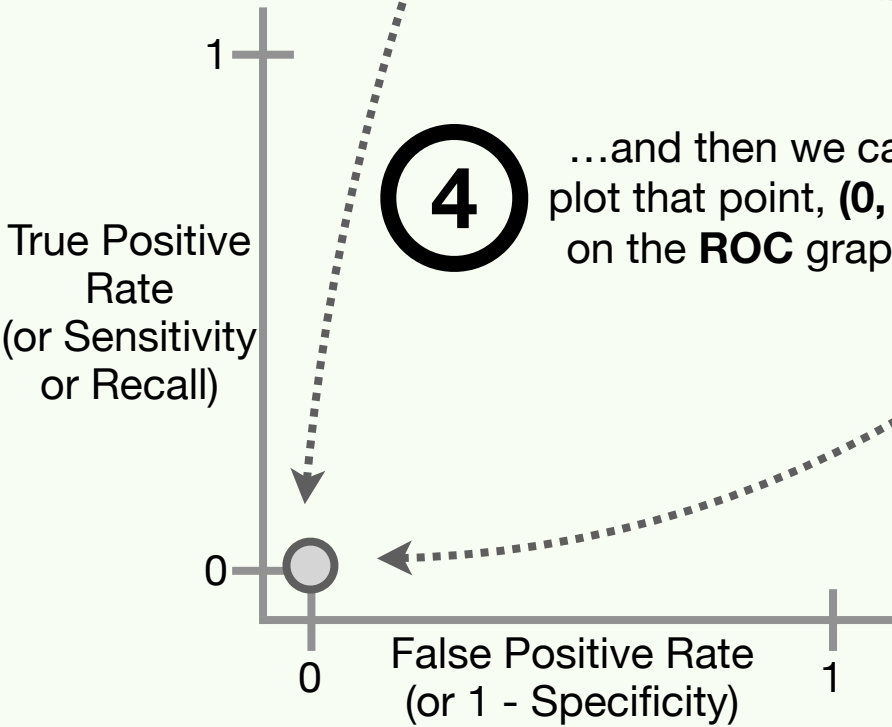
2 Using the values in the **Confusion Matrix**, we can calculate the **True Positive Rate**...

$$\text{True Positive Rate} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$
$$= \frac{0}{0 + 5} = 0$$

3 ...and the **False Positive Rate**...

$$\text{False Positive Rate} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$
$$= \frac{0}{0 + 4} = 0$$

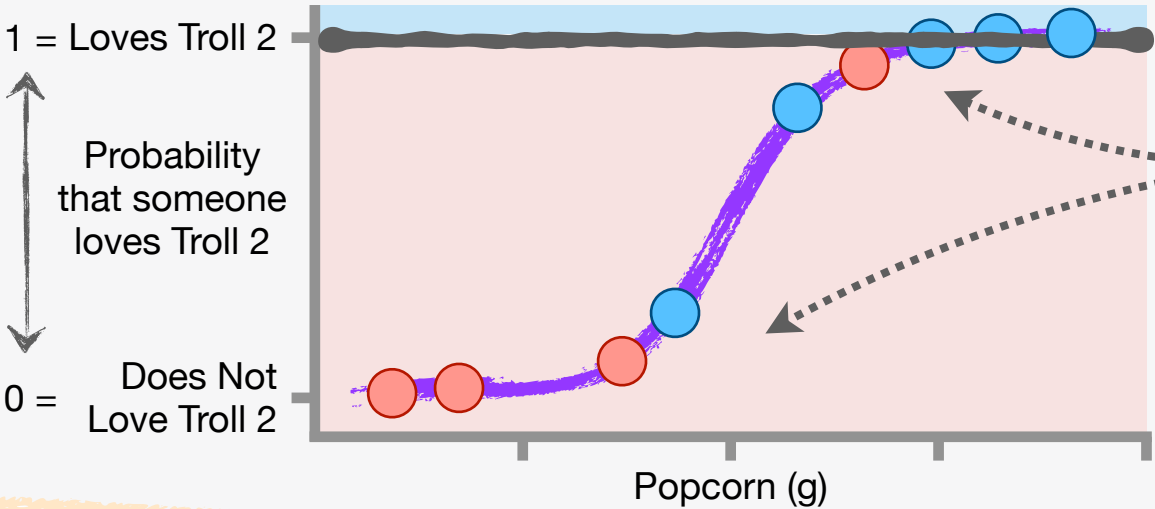
4 ...and then we can plot that point, **(0, 0)**, on the **ROC** graph.



ROC: Details Part 2

5 Now let's lower the classification threshold to **0.975**...

...which is just enough to classify one person as someone who **Loves Troll 2**...



Gentle Reminder:

	Predicted		
	Yes	No	
Actual	Yes	No	
	TP	FN	
	No	Yes	
	FP	TN	

True Positive

False Positive

False Negative

True Negative

...and everyone else is classified as someone who **Does Not Love Troll 2**...

...and that gives us this **Confusion Matrix**.

		Predicted	
		Yes	No
Actual	Yes	1	4
	No	0	4

Threshold = 0.975

6 Using the values in the **Confusion Matrix**, we can calculate the **True Positive Rate**...

$$\text{True Positive Rate} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

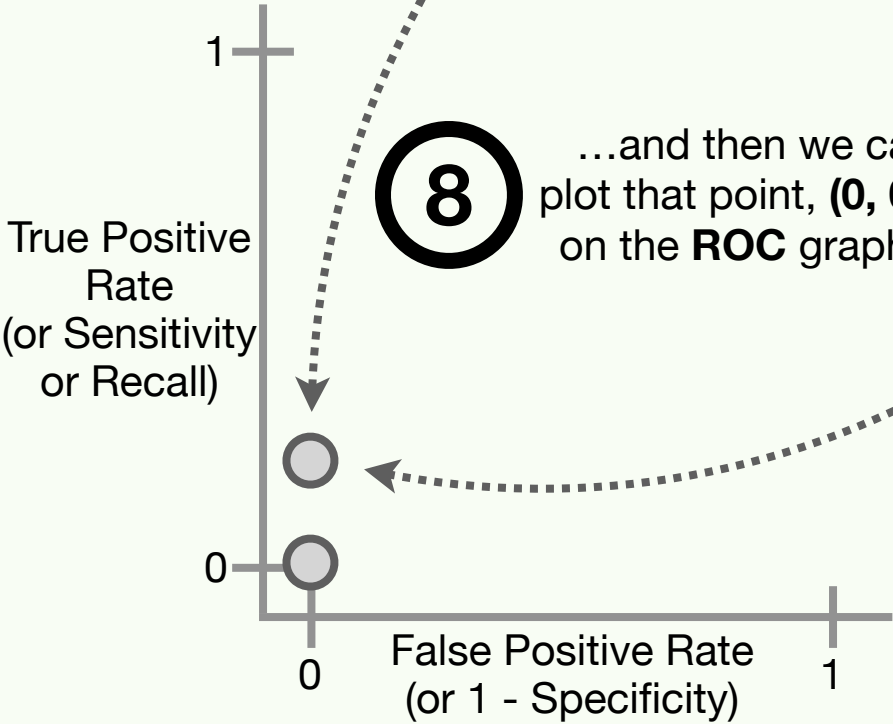
$$= \frac{1}{1 + 4} = 0.2$$

7 ...and the **False Positive Rate**...

$$\text{False Positive Rate} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$

$$= \frac{0}{0 + 4} = 0$$

8 ...and then we can plot that point, **(0, 0.2)**, on the **ROC** graph...

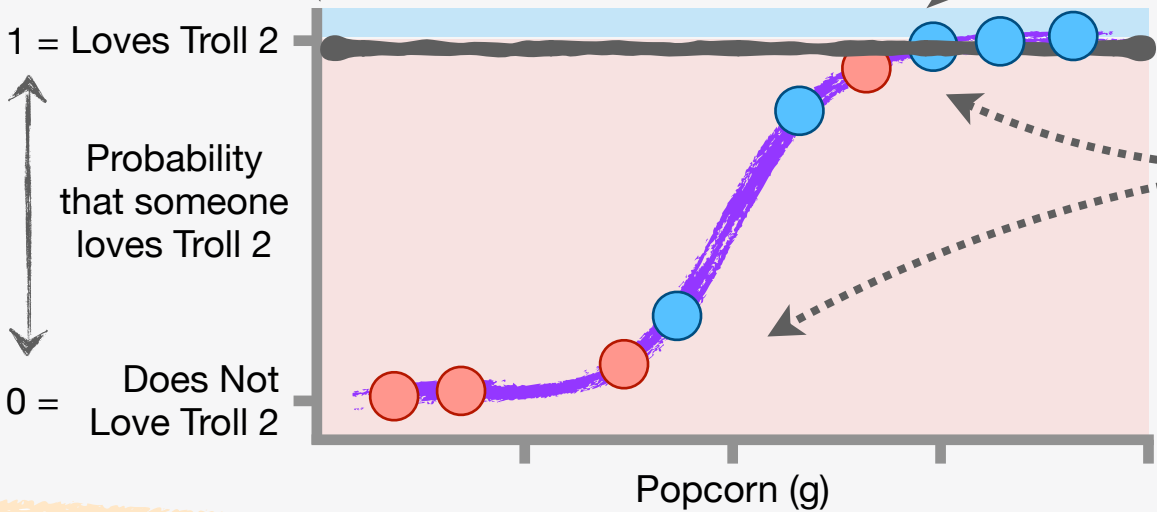


...and the new point is above the first point, showing that the new threshold increases the proportion of *actual Positives* that were *correctly* classified. **BAM!!!**

ROC: Details Part 3

9 Now let's lower the classification threshold to **0.965**...

...which is just enough to classify **2** people as people who **Love Troll 2**...



Gentle Reminder:

	Predicted		
	Yes	No	
Actual	Yes	FN	False Negative
	No	TN	
True Positive		True Negative	
False Positive			

...and everyone else is classified as someone who **Does Not Love Troll 2**...

...and that gives us this **Confusion Matrix**.

		Predicted	
		Yes	No
Actual	Yes	2	3
	No	0	4

Threshold = 0.965

10 Using the values in the **Confusion Matrix**, we can calculate the **True Positive Rate**...

True Positive Rate = $\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$

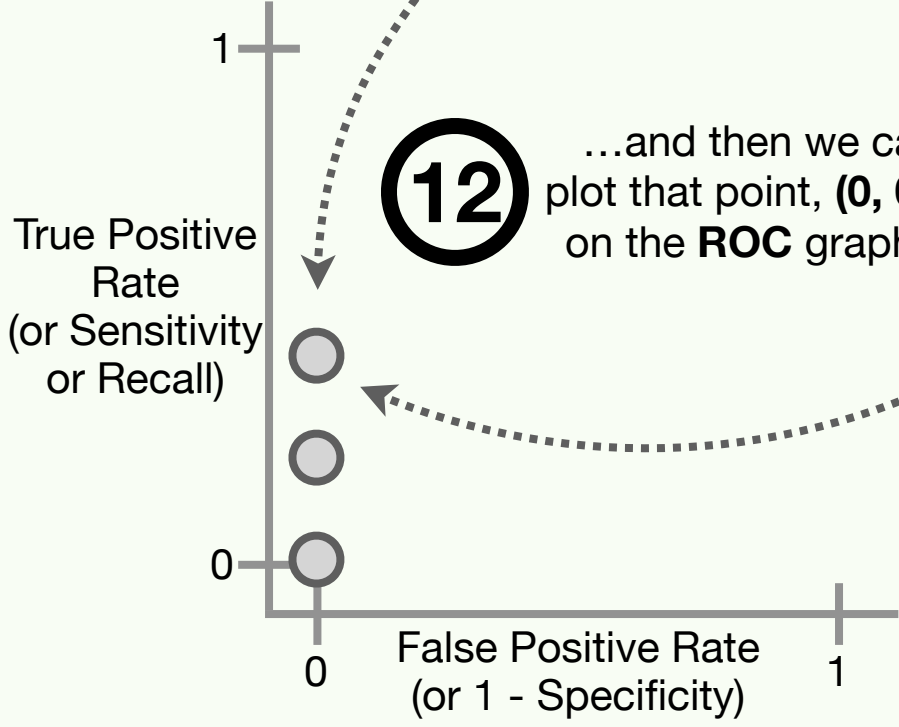
$= \frac{2}{2 + 3} = 0.4$

11 ...and the **False Positive Rate**...

False Positive Rate = $\frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$

$= \frac{0}{0 + 4} = 0$

12 ...and then we can plot that point, **(0, 0.4)**, on the **ROC** graph...

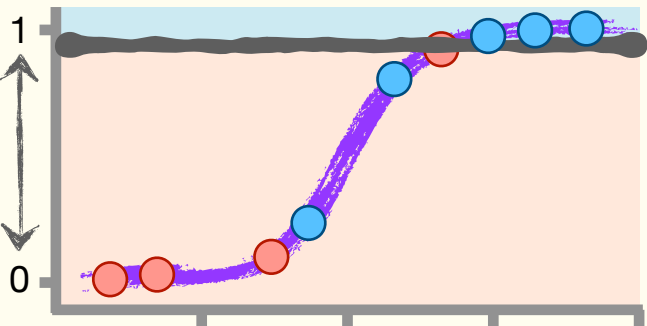


...and the new point is above the first two points, showing that the new threshold increases the proportion of *actual Positives* that were *correctly* classified.

ROC: Details Part 4

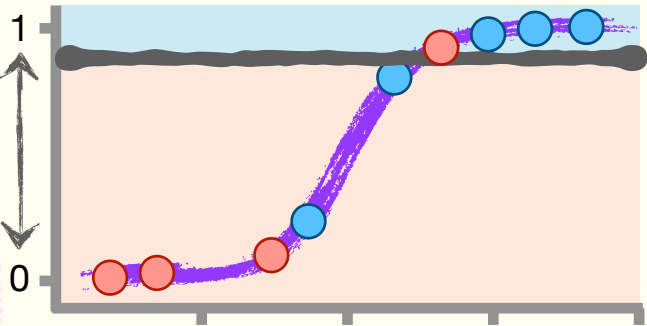
13

Likewise, for each threshold that increases the number of **Positive** classifications (in this example, that means classifying a person as someone who **Loves Troll 2**), we calculate the **True Positive Rate** and **False Positive Rate** until everyone is classified as **Positive**.



		Predicted	
		Yes	No
Actual	Yes	3	2
	No	0	4

Threshold = 0.95



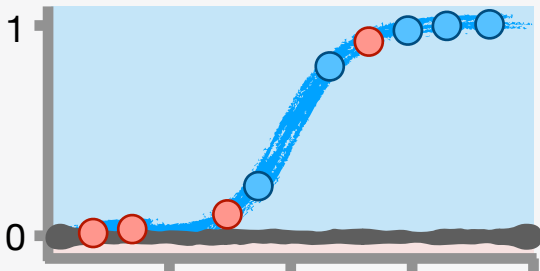
		Predicted	
		Yes	No
Actual	Yes	3	2
	No	1	3

Threshold = 0.87

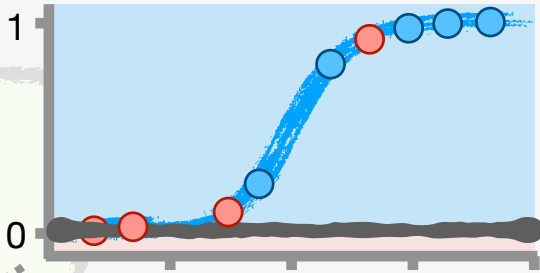
True Positive Rate (or Sensitivity or Recall)

False Positive Rate (or 1 - Specificity)

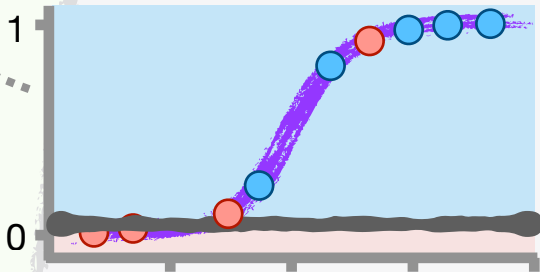
Threshold = 0



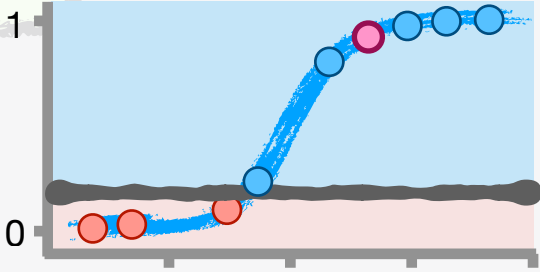
Threshold = 0.0645



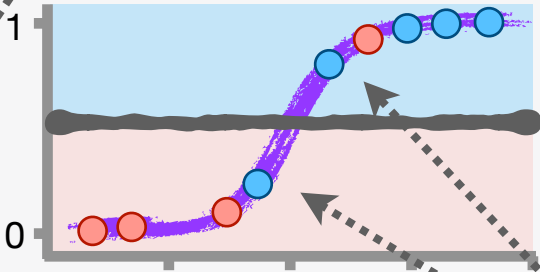
Threshold = 0.007



Threshold = 0.195



Threshold = 0.5



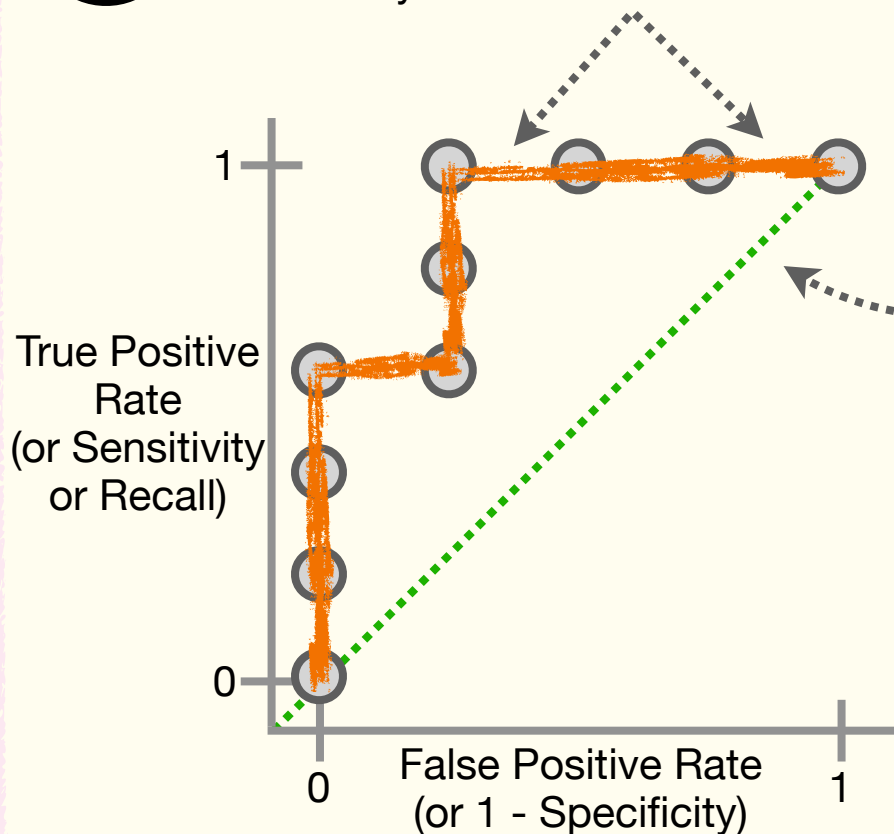
NOTE: Although there are a lot of potential thresholds between these two points, they all result in the same **True Positive Rate** and the same **False Positive Rate**, so it doesn't matter which one we pick, we just have to pick one of them.

ROC: Details Part 5

14

After we finish plotting the points from each possible **Confusion Matrix**, we usually **connect the dots**...

...and add a **diagonal line** that tells us when the **True Positive Rate = False Positive Rate**.

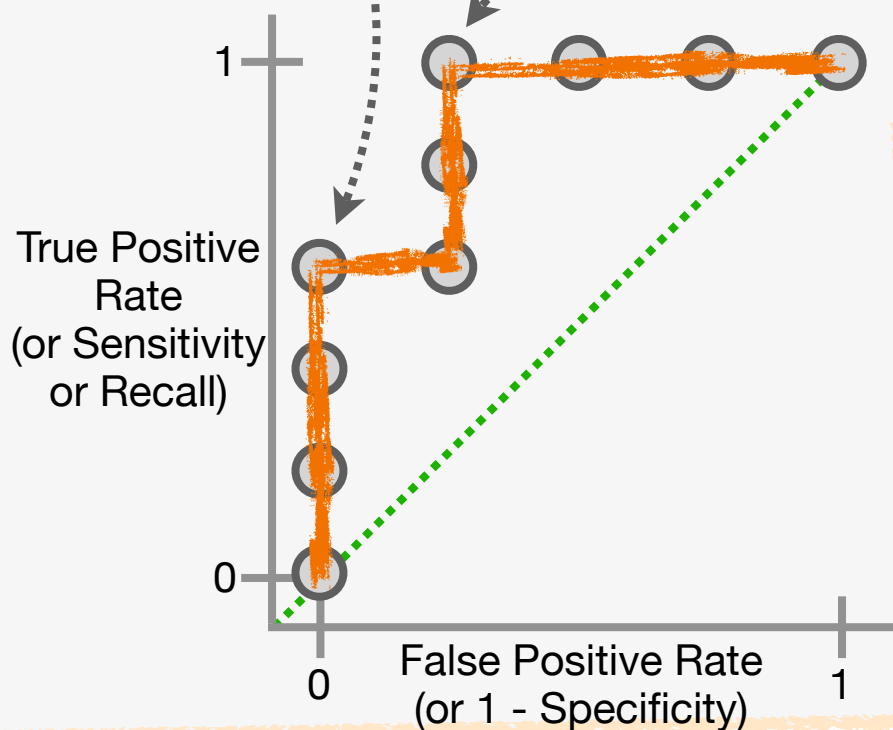


15

Now, without having to sort through a huge pile of **Confusion Matrices**, we can use the **ROC graph** to pick a classification threshold.

If we want to avoid all **False Positives**, but want to maximize the number of actual **Positives** correctly classified, we would pick this threshold...

...but if we can tolerate a few **False Positives**, we would pick this threshold because it *correctly* classifies all of the actual **Positives**.



16

ROC graphs are great for selecting an optimal classification threshold for a model. But what if we want to compare how one model performs vs. another? This is where the **AUC**, which stands for **Area Under the Curve**, comes in handy. So read on!!!

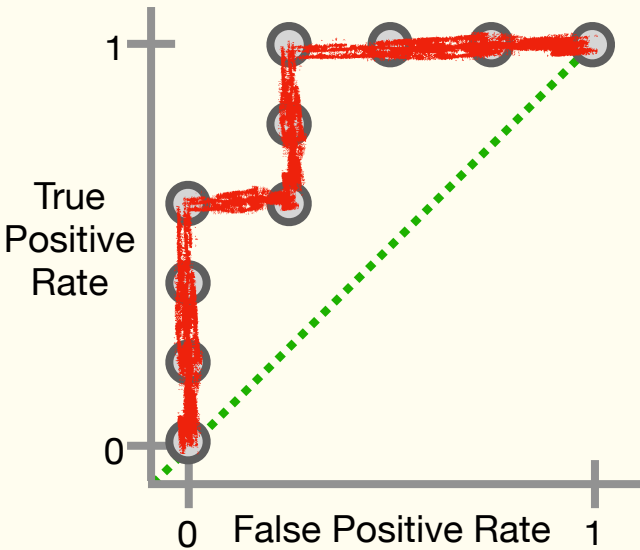
AUC: Main Ideas

1

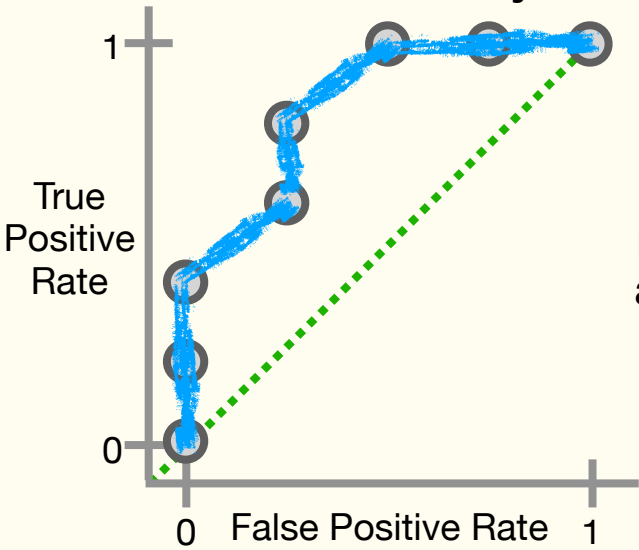
Now, imagine we created **Logistic Regression** and **Naive Bayes** models and tested them with the same data, and we wanted to know which model performed better.

In theory, we could compare the individual **ROC** graphs, and when we only have two models to compare, this is a pretty good option.

ROC for Logistic Regression



ROC for Naive Bayes



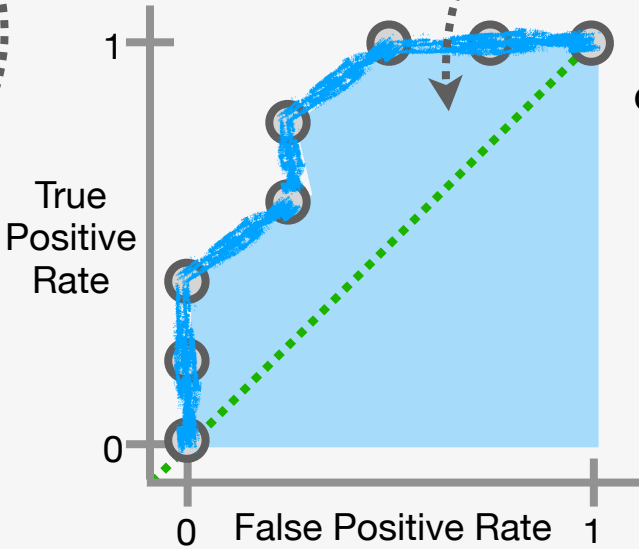
However, if we wanted to compare a bunch of models, this would be just as tedious as comparing a bunch of **Confusion Matrices**.
UGH!!!

2

So, instead of comparing a bunch of **ROC** graphs, one simple way to summarize them is to calculate and compare the **AUC**: the **Area Under** each **Curve**.

In this case, the **AUC** for **Logistic Regression** is 0.9...

...and the **AUC** for **Naive Bayes** is 0.85...



...and because **Logistic Regression** has a larger **AUC**, we can tell that, overall, **Logistic Regression** performs better than **Naive Bayes** with these data.